A Survey of Inverse Dynamics Models

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I. Related Work

World models learn environment dynamics for prediction and planning, ranging from latent dynamics models to large-scale video-based models trained on diverse sequential data, and offer a scalable framework for policy evaluation and optimization [1], [2]. To improve robustness under limited action supervision, recent works incorporate verification and inverse dynamics constraints to enforce action-consistent predictions and enable self-improvement [3]. Inverse dynamics models (IDMs), which infer actions from state transitions, they can be broadly categorized into explicit and implicit approaches.

A. Implicit Inverse Dynamics Modeling

Implicit inverse dynamics modeling captures the relationship between state transitions and actions without introducing an explicit action prediction module. Instead, actions are implicitly encoded within joint generative or sequential modeling of observations and controls.

Joint video–action modeling for implicit inverse dynamics. A representative line of work, by coupling video and action generation for end-to-end learning, enables predicted visual dynamics to improve latent representations, where inverse dynamics is implicitly captured to generate actions [4]–[15]. A representative example is DreamZero [16], which enables leveraging large-scale action-free data without relying on repetitive demonstrations, leading to improved generalization across diverse tasks and environments [16]. Despite these advantages, using DreamZero as a representative example, we discuss three potential limitations.

Alignment. Such implicit formulations rely on the alignment between generated visual dynamics and actions, which is not explicitly enforced. As noted in the paper, *"we observe that the predicted actions closely align with the generated video"* [16], suggesting that such alignment may emerge implicitly without explicit constraints, though it is primarily supported by qualitative observations. Although joint modeling can produce seemingly consistent behavior, it does not guarantee that visual dynamics and actions are truly aligned. The model may effectively learn to predict visual transitions and actions independently, without enforcing precise correspondence between them. As a result, alignment is only implicitly learned and can be fragile, leading to misaligned behavior when predicted dynamics deviate from reality.

IDM may be less robust. By coupling video and action generation, these approaches learn predictive representations from visual dynamics, effectively mapping 2D image transitions to underlying 3D actions. This introduces an inherently under-constrained problem, as 2D observations do not uniquely determine the corresponding action. While models such as DreamZero leverage strong video priors to mitigate this ambiguity, the inverse mapping remains ill-posed, might make the learned dynamics less robust, especially under novel viewpoints or distribution shifts.

Lack of verification and learning from prediction errors. During inference, DreamZero replaces predicted frames with ground-truth observations to prevent error accumulation in closed-loop control [16]. While this stabilizes execution, it also removes opportunities for the model to verify and learn from its own prediction errors.

Sequence-based implicit inverse dynamics. Another line of work adopts sequence modeling, where observations, actions, and rewards are tokenized and jointly modeled via next-token prediction, as exemplified by iVideoGPT [17]. In such approaches, action information is entangled within the sequence and inferred implicitly through autoregressive generation.

B. Explicit Inverse Dynamics Modeling

In contrast to implicit formulations, explicit inverse dynamics models introduce a dedicated action prediction pathway, where actions are explicitly inferred from state transitions.

Two-frame / video-based IDM. A common formulation predicts actions directly from consecutive observations or planned visual goals. Early works learn to infer actions from video transitions or video-conditioned goals, e.g., predicting actions from pairs of frames or future visual plans [18]–[25]. These approaches treat inverse dynamics as an explicit mapping from visual transitions to actions, but often do not fully exploit the underlying physical structure of the environment. Moreover, inferring 3D actions from 2D image observations is inherently under-constrained, which may limit robustness, especially under novel viewpoints or distribution shifts.

Optical flow-based IDM. Another line of work leverages dense correspondences such as optical flow to bridge visual changes and actions. For example, actions can be inferred from estimated pixel correspondences or motion fields [26], or through intermediate latent action representations derived from visual motion [27]. While these methods introduce stronger structure, they may still suffer from information loss.

Tool-centric IDM. Tool-centric approaches instead focus on modeling the motion of manipulators(e.g., hand) or tools as a proxy for action. Methods such as large video planners and video-driven policy learning explicitly track tool

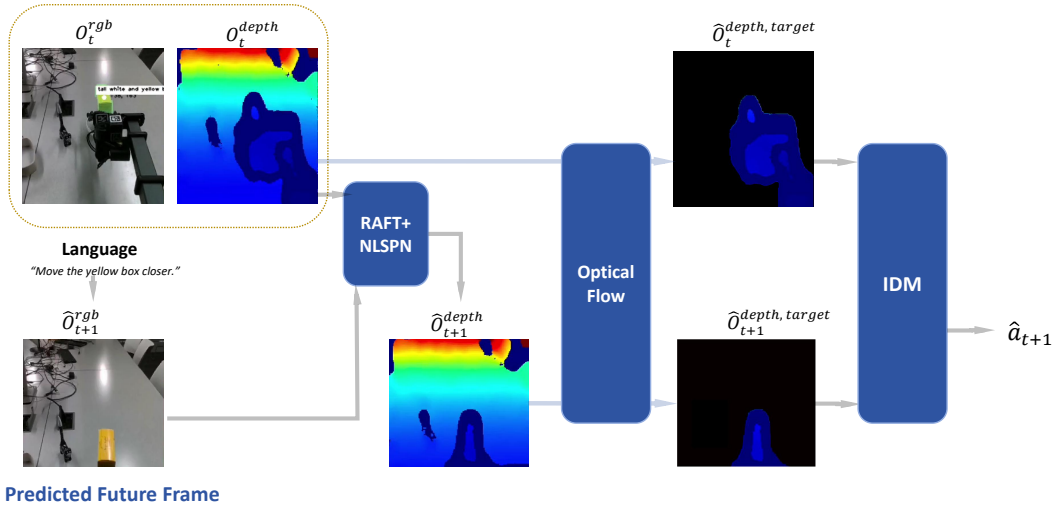

Fig. 1: Overview of the proposed IDM pipeline. Given the current RGB-D observation and a language instruction, a future frame is first predicted to represent the desired outcome. Depth is then propagated using optical flow (RAFT) and completed via NLSPN. Motion between frames is used to implicitly identify the manipulated object, yielding target-centric depth representations. Finally, an inverse dynamics model predicts the action that realizes the desired transition.

trajectories and map them to control signals [28], [29]. For example, Dreamitate [29] formulates policy learning as video generation followed by trajectory extraction and execution. Specifically, the model first generates a video of task execution conditioned on the current observation, from which a 2D trajectory of the tool is extracted. This trajectory is then lifted to 3D space by matching it to feasible trajectories via MegaPose, which is used to recover accurate end-effector states from RGB observations, then actions are obtained via inverse kinematics. The availability of 3D information (e.g., depth) enables accurate trajectory execution for fine-grained manipulation tasks. This also leads to strong generalization across unseen objects, environments, and embodiments, as the policy focuses on consistent tool motion.

However, such approaches, instead of learning the causal relationship between actions and object state transitions, they primarily track and execute tool trajectories (e.g., end-effector motion), making them closer to imitation learning of trajectories rather than true dynamics modeling. This introduces a structural mismatch: successful execution relies on the assumption that tool motion faithfully reflects object interaction, which may not always hold in complex environments. As a result, these methods may fail to capture the underlying action–object causality that is central to world modeling.

3D and object-centric IDM. More structured formulations further incorporate 3D information or object-centric representations. For instance, some methods recover 3D states or poses from visual observations and perform inverse kinematics to obtain actions [29], while others learn dense object descriptors or keypoint-based representations to explicitly construct inverse mappings [30], [31]. These approaches enables precise control and fine-grained manipulation. However, they often require additional assumptions on object structure or access to reliable 3D information.

Latent representation IDM. Finally, recent works explore learning inverse dynamics in latent representation spaces, where actions are predicted from learned visual embeddings [32]–[35]. While scalable, such approaches depend on the quality of learned representations, typically produced by general-purpose encoders (e.g., ViT or CNN), which may lack precise spatial grounding for accurate control.

Overall, existing inverse dynamics models often exhibit limited robustness, partly due to insufficient spatial information for accurate action grounding.

II. Potential Directions

Action–image consistency for joint video–action modeling. A potential direction is to improve alignment in joint video–action modeling for implicit inverse dynamics by explicitly enforcing consistency between predicted visual dynamics and generated actions.

Error-driven model verification and refinement. A potential direction is to leverage execution-time prediction errors as a learning signal to verify and update the model, rather than discarding them through ground-truth replacement.

Spatially grounded inverse dynamics for precise manipulation and improved generalization. We detail the proposed approach in the following section.

III. Method of the Proposed IDM

We propose a **Spatially Grounded IDM** for **Precise Manipulation and Improved Generalization**(Fig 1).

Given the current observation and a language instruction, we first predict a future frame that reflects the desired outcome. Depth maps are then estimated for the predicted frame. We further extract target-centric depth using optical flow, where **the manipulated object is implicitly defined as regions exhibiting motion across frames**. Finally, an

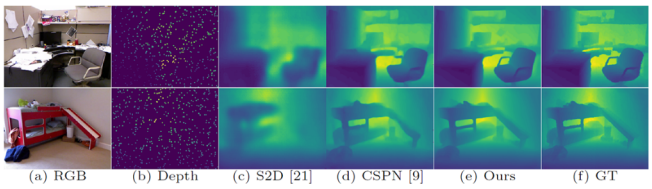

Fig. 2: Example of depth completion using NLSPN

inverse dynamics model takes the resulting geometric transformation as input and predicts the action that realizes the transition.

A. Pipeline detail

Future Frame Prediction. We first predict a future observation $\hat{O}_{t+1}^{rgb}$ conditioned on the current observation O_t^{rgb} and the language instruction. This predicted frame represents the desired target state implied by the instruction.

Future Depth Estimation. Given the current RGB-D observation (O_t^{rgb}, O_t^{depth}) and the predicted future RGB frame $\hat{O}_{t+1}^{rgb}$, the goal is to estimate the corresponding future depth $\hat{O}_{t+1}^{depth}$. This can be achieved through several possible pipelines. Here, we adopt a flow-based depth propagation approach using RAFT for optical flow estimation and NLSPN for depth completion.

Specifically, we first estimate optical flow between O_t^{rgb} and $\hat{O}_{t+1}^{rgb}$ using RAFT to obtain pixel correspondences. The depth map O_t^{depth} is then warped to the target frame using the estimated flow, producing a sparse depth map $\tilde{O}_{t+1}^{depth}$. Due to occlusions and matching errors, the warped depth is incomplete. We then apply NLSPN to recover a dense depth map $\hat{O}_{t+1}^{depth}$ (Fig. 2).

Target Extraction via Motion. We extract target-centric depth using optical flow computed between O_t^{rgb} and $\hat{O}_{t+1}^{rgb}$. Specifically, the manipulated object is implicitly defined as regions exhibiting motion across frames. Applying this motion mask to the depth maps yields target-focused depth representations, denoted as $O_t^{depth,\text{target}}$ and $\hat{O}_{t+1}^{depth,\text{target}}$.

Inverse Dynamics Prediction. Finally, an inverse dynamics model takes the geometric transformation as input and predicts the action a_{t+1} that realizes the transition from $O_t^{depth,\text{target}}$ to $\hat{O}_{t+1}^{depth,\text{target}}$.

B. Advantages

Our pipeline offers several desirable properties. **Strong generalization through geometry.** By relying on spatially grounded geometric representations rather than appearance, the model naturally generalizes across variations in background, object appearance, and lighting conditions. Since the inverse dynamics model operates on target-centric depth, irrelevant visual details are abstracted away, retaining only essential geometric and spatial relationships. This also facilitates generalization across tasks that share similar subgoals, even when the overall task differs.

Accurate and fine-grained manipulation. The use of explicit 3D information enables more precise spatial reasoning and supports fine manipulation. Compared to purely image-based representations, depth-based geometric cues improve robustness under viewpoint changes and allow accurate execution of detailed actions.

Implicit target identification via motion. The manipulated object is implicitly identified through motion, eliminating the need for explicit object detection or segmentation (e.g., SAM), and enabling a more flexible and scalable formulation.

Simplified and learnable inverse dynamics. The resulting inverse dynamics problem becomes significantly simpler, as the input captures only the essential geometric changes, without requiring the model to fit irrelevant visual details. This abstraction improves generalization while reducing model complexity, making it akin to classical control or inverse kinematics. As a result, the formulation enhances both learnability and data efficiency.

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